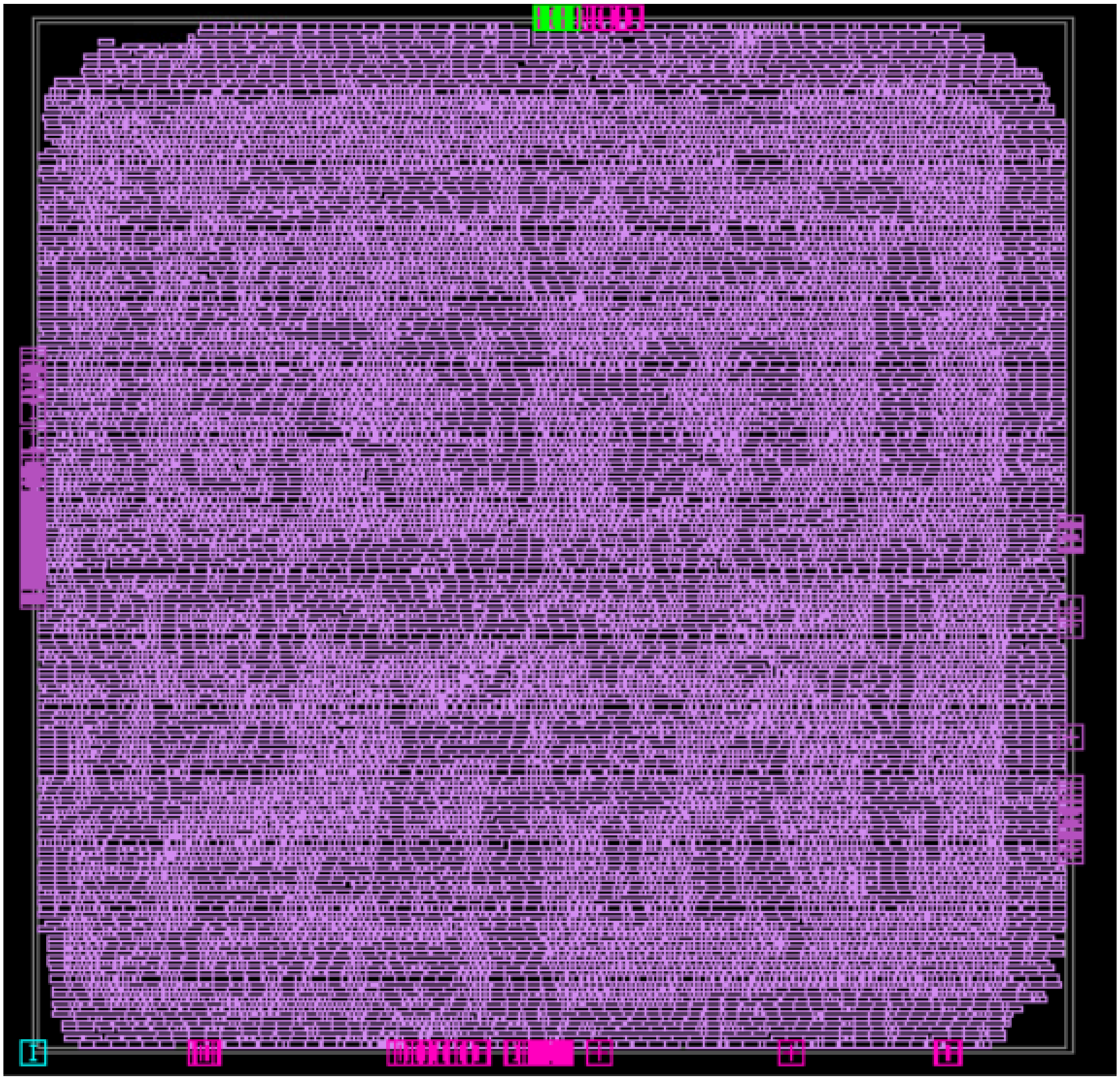
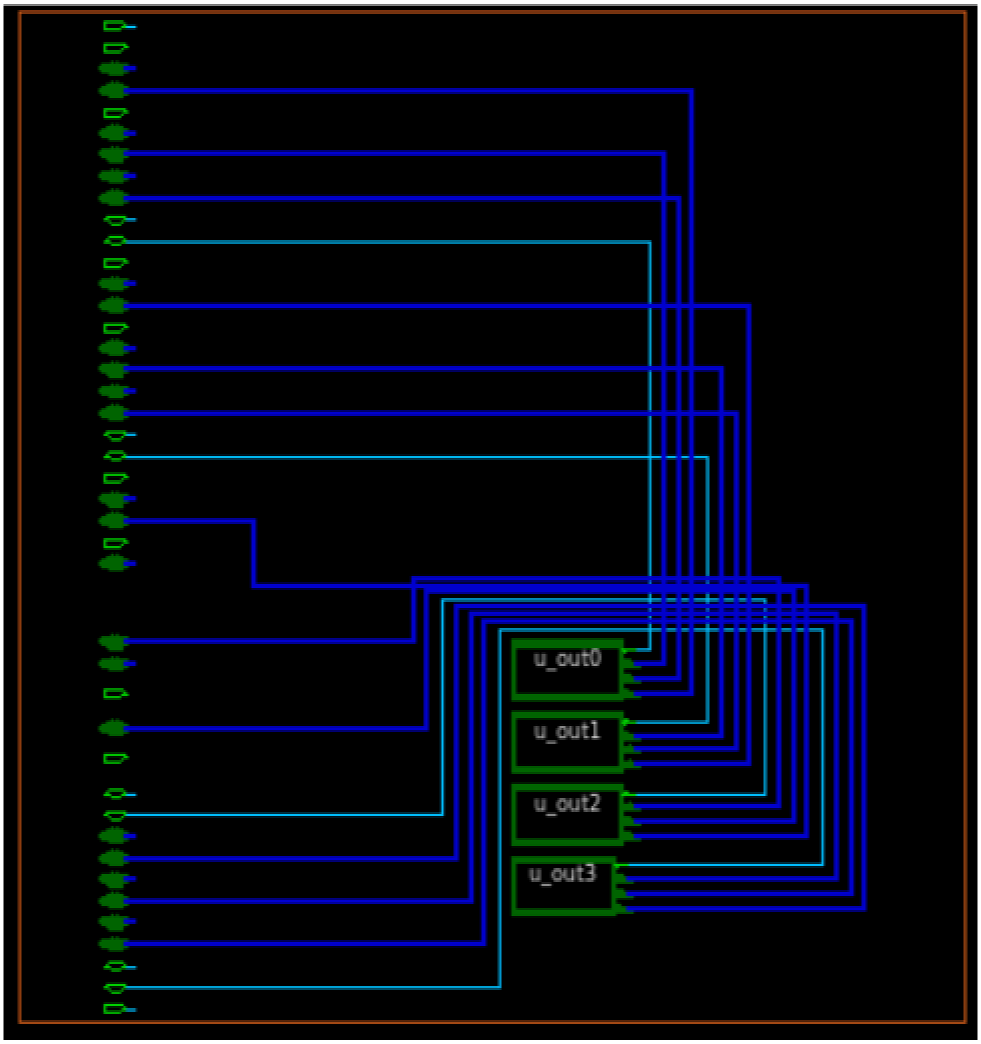
With clock gating

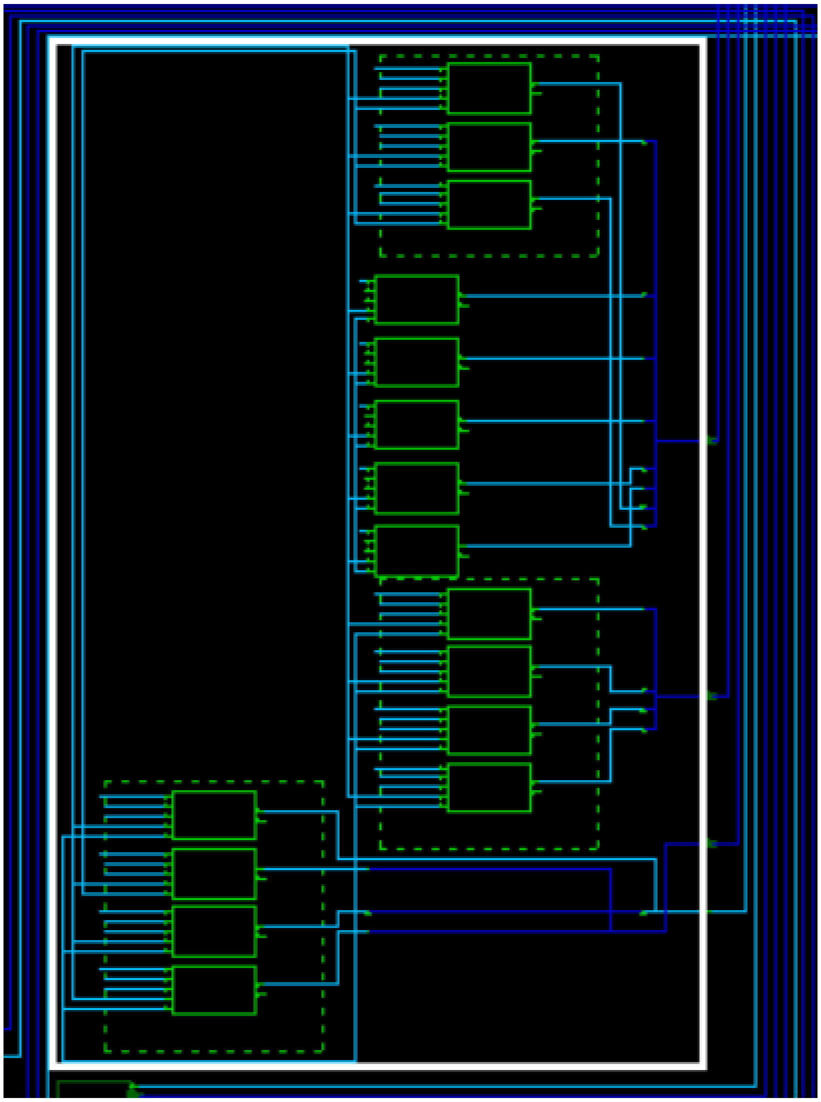


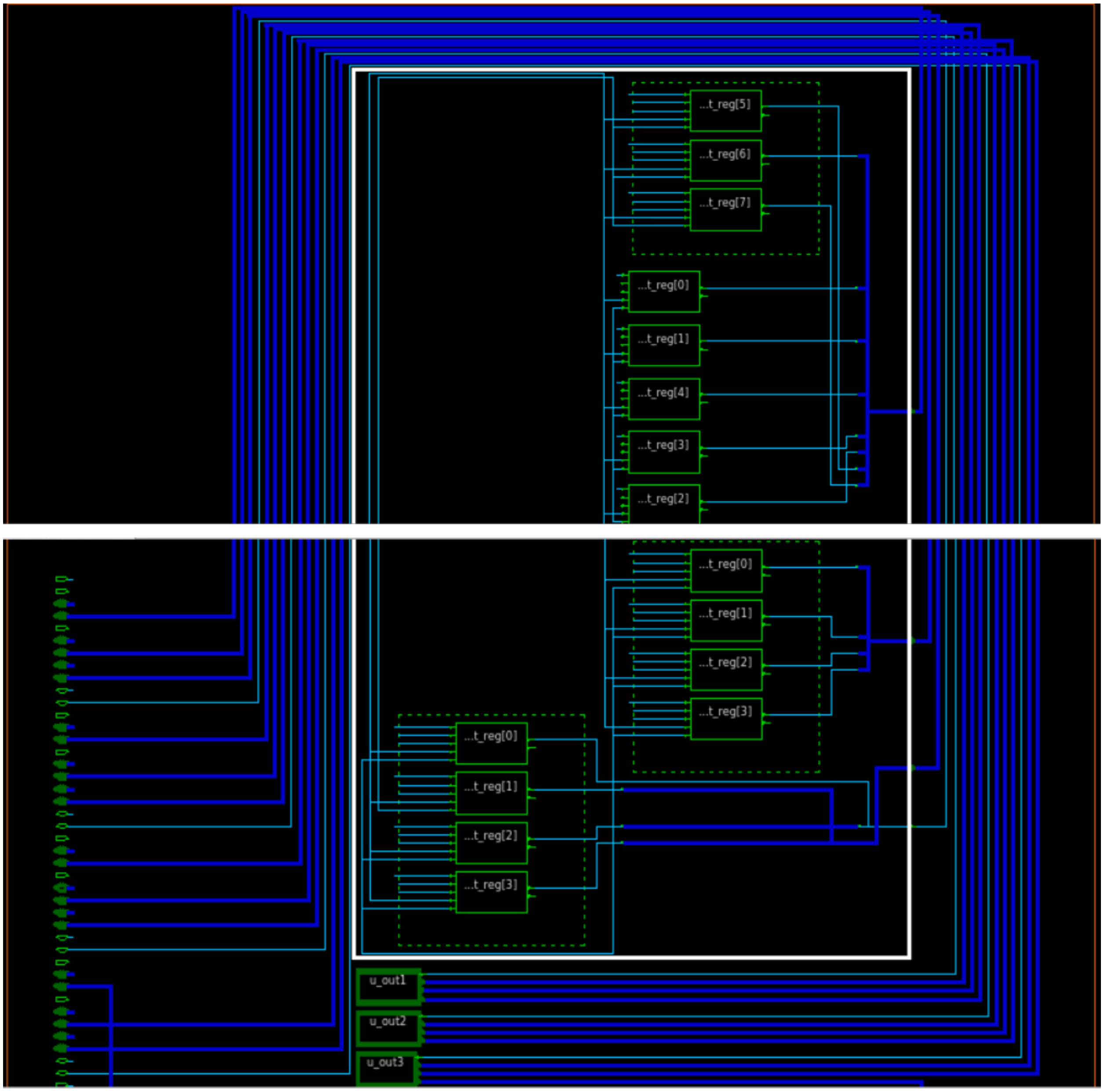
Schematic



u_out0
u_out1
u_out2
u_out3

Inside each port





Gate level netlist is attached as switch_4port.v

Timing report

fc_shell> report_timing
****
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -report_by design
Design : switch_4port
Version: V-2023.12-SP3
Date   : Sun Jan 18 15:15:03 2026

```
****************************************

 Startpoint: count_reg[3][4] (rising edge-triggered flip-flop clocked by clk)
 Endpoint: clock_gate_fifo_reg_227 (rising clock gating-check end-point clocked by clk)
 Mode: FUNC
 Corner: Slow
 Scenario: FUNC_Slow
 Path Group: clk
 Path Type: max

 Point                                            Incr      Path
 ------------------------------------------------------------------------
 clock clk (rise edge)                            0.00      0.00
 clock network delay (ideal)                     -0.02     -0.02

 count_reg[3][4]/CLK (SDFFARX1_HVT)               0.00     -0.02 r
 count_reg[3][4]/Q (SDFFARX1_HVT)                 0.65      0.63 r
 ctmi_4173/Y (NOR2X1_RVT)                         0.19      0.83 f
 phfnr_buf_8188/Y (INVX1_HVT)                     0.13      0.96 r
 ctmi_82644/Y (OR3X1_RVT)                         0.19      1.14 r
 ctmTdsLR_1_11466/Y (NAND2X0_RVT)                 0.09      1.23 f
 ctmTdsLR_2_11467/Y (AND2X1_RVT)                  0.17      1.40 f
 ctmi_82642/Y (NAND2X0_RVT)                       0.13      1.54 r
 phfnr_buf_8221/Y (INVX1_HVT)                     0.31      1.84 f
 ctmi_82641/Y (NAND3X0_RVT)                       0.29      2.14 r
 phfnr_buf_8232/Y (INVX1_HVT)                     0.38      2.52 f
 ctmi_82659/Y (AND2X1_RVT)                        0.34      2.87 f
 ctmi_82658/Y (AND2X1_RVT)                        0.21      3.08 f
 ctmi_82660/Y (NAND2X0_RVT)                       0.18      3.26 r
 ctmi_82657/Y (OA21X2_HVT)                        1.46      4.71 r
 ctmi_82744/Y (NAND2X0_RVT)                       0.18      4.89 f
 ctmi_82742/Y (NOR4X1_RVT)                        0.29      5.18 r
 ctmi_9092/Y (OA221X1_RVT)                        0.17      5.36 r
 ctmi_82739/Y (AO22X2_RVT)                        0.22      5.58 r
 HFSINV_653_9890/Y (INVX2_RVT)                    0.10      5.69 f
 ctmi_82769/Y (AND2X2_LVT)                        0.14      5.83 f
 ctmi_82768/Y (AND2X1_RVT)                        0.18      6.01 f
 ctmi_4182/Y (NAND2X4_RVT)                        0.24      6.25 r
 HFSINV_2391_11005/Y (INVX4_RVT)                  0.12      6.37 f
 ctmi_82952/Y (OA21X1_RVT)                        0.20      6.57 f
 HFSINV_40_9775/Y (INVX0_HVT)                     0.16      6.73 r
 ctmi_82951/Y (AND2X2_RVT)                        0.26      6.99 r
 ctmi_82950/Y (AND2X2_RVT)                        0.25      7.25 r
 ctmi_83009/Y (NAND2X4_HVT)                       1.33      8.57 f
 HFSINV_816_10087/Y (INVX2_RVT)                   0.35      8.93 r
 ctmi_83004/Y (AO222X1_RVT)                       0.33      9.26 r
 clock_gate_fifo_reg_227/EN (CGLPPRX2_HVT)        0.00      9.26 r
 data arrival time                                          9.26

 clock clk (rise edge)                           10.00     10.00
 clock network delay (ideal)                     -0.36      9.64
 clock_gate_fifo_reg_227/CLK (CGLPPRX2_HVT)       0.00      9.64 r
 clock uncertainty                               -0.15      9.49
 library setup time                              -0.22      9.26
 data required time                                         9.26
 ------------------------------------------------------------------------
 data required time                                         9.26
 data arrival time                                         -9.26
 ------------------------------------------------------------------------
 slack (MET)                                                0.00
```

1

Power report

```
fc_shell> report_power
****************************************
Report : power
        -significant_digits 2
Design : switch_4port
Version: V-2023.12-SP3
Date   : Sun Jan 18 15:15:49 2026
****************************************
Information: Activity propagation will be performed for scenario FUNC_Slow.
Information: Doing activity propagation for mode 'FUNC' and corner 'Slow' with effort level 'medium'. (POW-024)
Information: Timer-derived activity data is cached on scenario FUNC_Slow (POW-052)
Infomation: Fast mode activity propagation power.rtl_activity_annotation setup is ignored. Always use accurate mode.
Information: Running switching activity propagation with 4 threads!

 **** Information : No. of simulation cycles = 6 ****
Mode: FUNC
Corner: Slow
Scenario: FUNC_Slow
Voltage: 0.75
Temperature: 125.00

Voltage Unit         : 1V
Capacitance Unit     : 1fF
Time Unit            : 1ns
Temperature Unit     : 1C
Dynamic Power Unit   : 1pW
Leakage Power Unit   : 1pW

Switched supply net power scaling:
scaling for leakage power

Supply nets:
VDD (power) probability 1.00 (default)
VSS (ground) probability 1.00 (default)
Warning: Fall toggles on pin count_reg[0][6]/QN are impossible given input states; converted to rise toggles. (POW-069)
Warning: Fall toggles on pin count_reg[1][6]/QN are impossible given input states; converted to rise toggles. (POW-069)
Warning: Fall toggles on pin count_reg[3][6]/QN are impossible given input states; converted to rise toggles. (POW-069)

  Cell Internal Power    = 4.15e+08 pW ( 77.6%)
  Net Switching Power    = 1.20e+08 pW ( 22.4%)
Total Dynamic Power      = 5.34e+08 pW (100.0%)

Cell Leakage Power       = 1.50e+09 pW


  Attributes
  ----------
      u  -  User defined power group
      i  -  Includes clock pin internal power

Power Group       Internal Power     Switching Power      Leakage Power        Total Power   (  %  )   Attrs
--------------------------------------------------------------------------------------------------------------------------
io_pad              0.00e+00           0.00e+00           0.00e+00           0.00e+00    (  0.0%)
memory              0.00e+00           0.00e+00           0.00e+00           0.00e+00    (  0.0%)
```

```
black_box          0.00e+00      0.00e+00      0.00e+00      0.00e+00  (  0.0%)
clock_network      1.19e+08      6.07e+06      1.30e+07      1.38e+08  (  6.8%)   i
register           1.64e+08      3.02e+06      7.92e+08      9.60e+08  ( 47.2%)
sequential         0.00e+00      0.00e+00      0.00e+00      0.00e+00  (  0.0%)
combinational      1.32e+08      1.10e+08      6.92e+08      9.34e+08  ( 46.0%)
-------------------------------------------------------------------------------------------------------------------------------
Total              4.15e+08 pW   1.20e+08 pW   1.50e+09 pW   2.03e+09 pW
1
```

Area report

```
fc_shell> report_area
Information: The command 'report_area' cleared the undo history. (UNDO-016)
****************************************
Report : area
Design : switch_4port
Version: V-2023.12-SP3
Date   : Sun Jan 18 15:16:39 2026
****************************************

Number of ports:                 307
Number of nets:                21258
Number of cells:               20977
Number of combinational cells: 16723
Number of sequential cells:     4254
Number of macros/black boxes:      0
Number of buf/inv:              1539
Number of references:            112

Combinational area:         47627.09
Buf/Inv area:                3403.50
Noncombinational area:      36067.61
Macro/Black Box area:           0.00

Total cell area:            83694.70
1
```

Utilization report

```
fc_shell> report_utilization
****************************************
Report : report_utilization
Design : switch_4port
Version: V-2023.12-SP3
Date   : Sun Jan 18 15:17:18 2026
****************************************
Utilization Ratio:              0.7239
Utilization options:
 - Area calculation based on:        site_row of block switch_4port
 - Categories of objects excluded:     hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                    115615.9509
Total Capacity Area:           115615.9509
Total Area of cells:           83694.7021
Area of excluded objects:
 - hard_macros      :          0.0000
 - macro_keepouts   :          0.0000
 - soft_macros      :          0.0000
 - io_cells         :          0.0000
 - hard_blockages   :          0.0000
```

Total Area of excluded objects: 0.0000
Ratio of excluded objects: 0.0000

Utilization of site-rows with:
 - Site 'unit':   0.7239

0.7239

Clock gating report

fc_shell> report_clock_gating
Information: The stitching and editing of coupling caps is turned OFF for design 'switch1.dlib:switch_4port.design'. (TIM-125)
Information: Design Average RC for design switch_4port (NEX-011)
Information: r = 1.051407 ohm/um, via_r = 0.460714 ohm/cut, c = 0.078705 ff/um, cc = 0.000000 ff/um (X dir) (NEX-017)
Information: r = 1.384202 ohm/um, via_r = 0.578659 ohm/cut, c = 0.091390 ff/um, cc = 0.000000 ff/um (Y dir) (NEX-017)
Information: The RC mode used is VR for design 'switch_4port'. (NEX-022)
Information: Update timing completed net estimation for all the timing graph nets (TIM-111)
Information: Net estimation statistics: timing graph nets = 21101, routed nets = 0, across physical hierarchy nets = 0, parasitics cached nets = 21101, delay annotated nets = 0, parasitics annotated nets = 0, multi-voltage nets = 0. (TIM-112)

```
****************************************
Report : clock_gating
Design : switch_4port
Version: V-2023.12-SP3
Date   : Sun Jan 18 15:57:44 2026
****************************************


                Clock Gating Summary
    ------------------------------------------------------------------
    |  Number of Clock gating elements        |         274       |
    |                                         |                   |
    |  Number of Tool-Inserted Clock gates    |      274 (100.00%) |
    |                                         |                   |
    |  Number of Pre-Existing Clock gates     |        0   (0.00%) |
    |                                         |                   |
    |  Number of Gated registers              |     3906  (98.14%) |
    |                                         |                   |
    |  Number of Tool-Inserted Gated registers |     3906  (98.14%) |
    |                                         |                   |
    |  Number of Pre-Existing Gated registers |        0   (0.00%) |
    |                                         |                   |
    |  Number of Ungated registers            |       74   (1.86%) |
    |                                         |                   |
    |  Total number of registers              |        3980       |
    |                                         |                   |
    |  Max number of Clock Gate Levels        |           2       |
    ------------------------------------------------------------------


1
```

Synthesis script

Switch.sdc

```
 # remove any constraints that were applied from previous runs:

remove_sdc -design
```

```
# Set the physical library:

set lib_name saed32hvt_ss0p75v125c


#set_current_design TOP


# Create clock object and set uncertainty

create_clock -period 10 [get_ports clk]

set_clock_uncertainty -setup 0.15 [get_clocks clk]


# Set constraints on input ports

#suppress_message UID-401

#suppress_message DCT-306


set_input_delay -max 0.5 -clock clk [remove_from_collection [all_inputs] [get_ports clk]]


#Set constraints on output ports:

set_output_delay 0.2 -max -clock clk [all_outputs]
```

Run.tcl

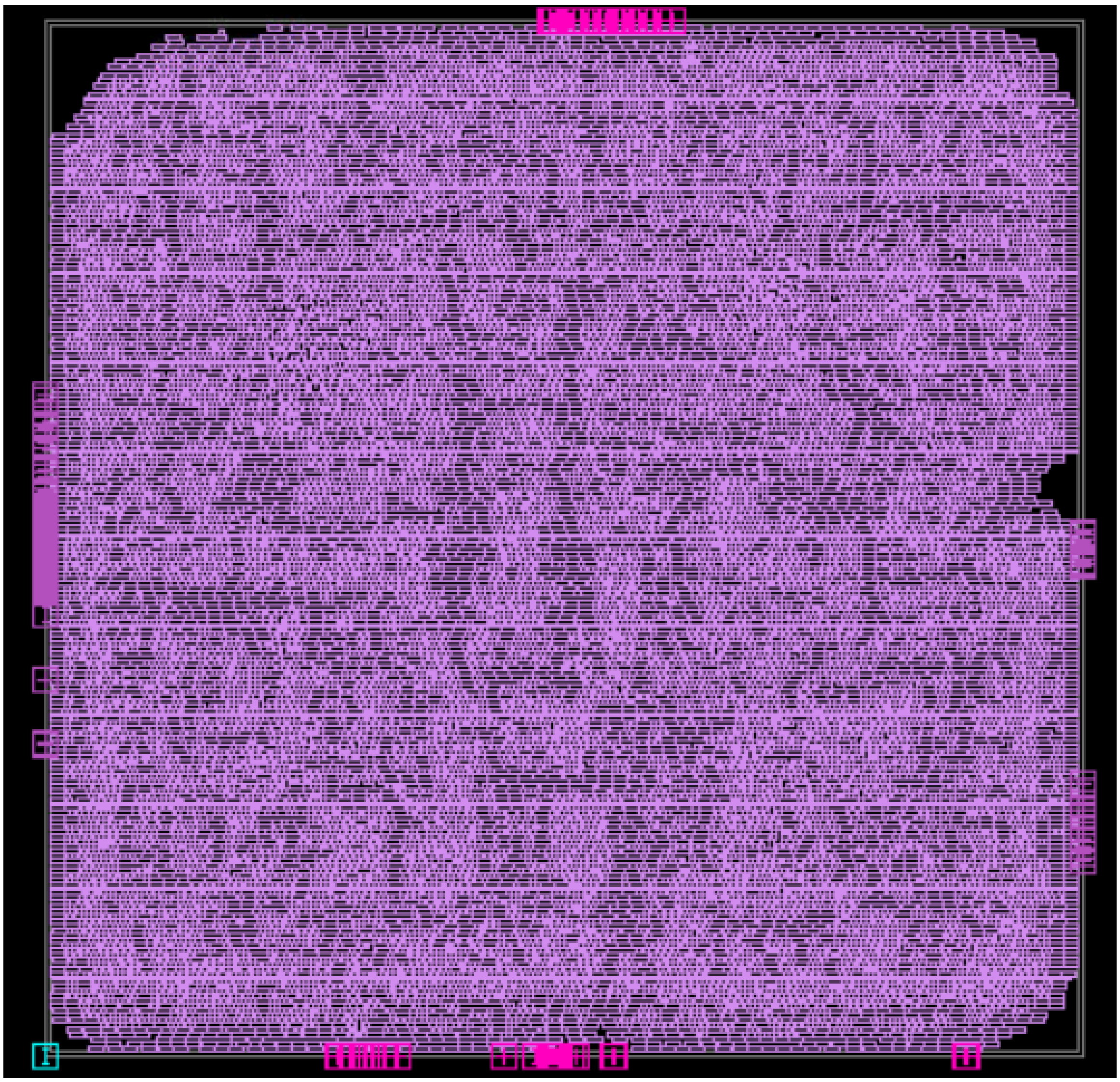
```
#lappend search_path scripts design_data
set_host_options -max_cores 4
set TECH_FILE     "/data/synopsys/lib/saed32nm/ref/tech/saed32nm_1p9m.tf"
##########################################################################
### Physical Library Settings
##########################################################################
create_lib  -technology $TECH_FILE  -ref_libs { /data/synopsys/lib/saed32nm/ref/CLIBs/saed32_hvt.ndm
/data/synopsys/lib/saed32nm/ref/CLIBs/saed32_lvt.ndm /data/synopsys/lib/saed32nm/ref/CLIBs/saed32_rvt.ndm }  switch1.dlib
open_lib switch1.dlib
report_ref_libs
read_parasitic_tech -tlu /data/synopsys/lib/saed32nm/ref/tech/saed32nm_1p9m_Cmax.lv.tluplus -name Cmax
read_parasitic_tech -tlu /data/synopsys/lib/saed32nm/ref/tech/saed32nm_1p9m_Cmin.lv.tluplus -name Cmin
#
save_lib
analyze -format sverilog {port_if.sv switch_port.sv switch_4port.sv}
elaborate switch_4port
set_top_module switch_4port
# save hir
set_app_options -name  compile.flow.autoungroup  -value false
#start_gui
save_block -as switch1/elaborate

# mcmm_setup:
# Remove all MCMM related info
remove_corners   -all
remove_modes     -all
remove_scenarios -all
# Create Corners
create_corner Fast
create_corner Slow
```

```
#
## Set parasitics parameters
set_parasitics_parameters -early_spec Cmin -late_spec  Cmin -corners {Fast}
set_parasitics_parameters -early_spec Cmax -late_spec  Cmax -corners {Slow}
#
## Create Mode
create_mode  FUNC
current_mode FUNC
#
## Create Scenarios
create_scenario -mode FUNC -corner Fast    -name FUNC_Fast
create_scenario -mode FUNC -corner Slow    -name FUNC_Slow
#

#sourse ConFiles/con431.con
current_scenario FUNC_Fast
source  /project/verif/users/mariandawud/ws/portsC/switch.sdc
current_scenario FUNC_Slow
source  /project/verif/users/mariandawud/ws/portsC/switch.sdc

set_auto_floorplan_constraints -core_utilization 0.7 -side_ratio {1 1} -core_offset 2

compile_fusion -to logic_opto
#create_placement
#legalize_placement
##Power
compile_fusion -to final_opto

save_block -as switch1/final_opto
report_timing

#set_attribute [get_layers {M1 M3 M5 M7 M9}]   routing_direction vertical
#set_attribute [get_layers {M2 M4 M6 M8 MRDL}] routing_direction horizontal

#source create_pg_network.tcl
#clock_opt
#route_opt
#save_block -as ALU/route_opt
write_sdf switch1.sdf
Write_verilog switch_4port.v
```

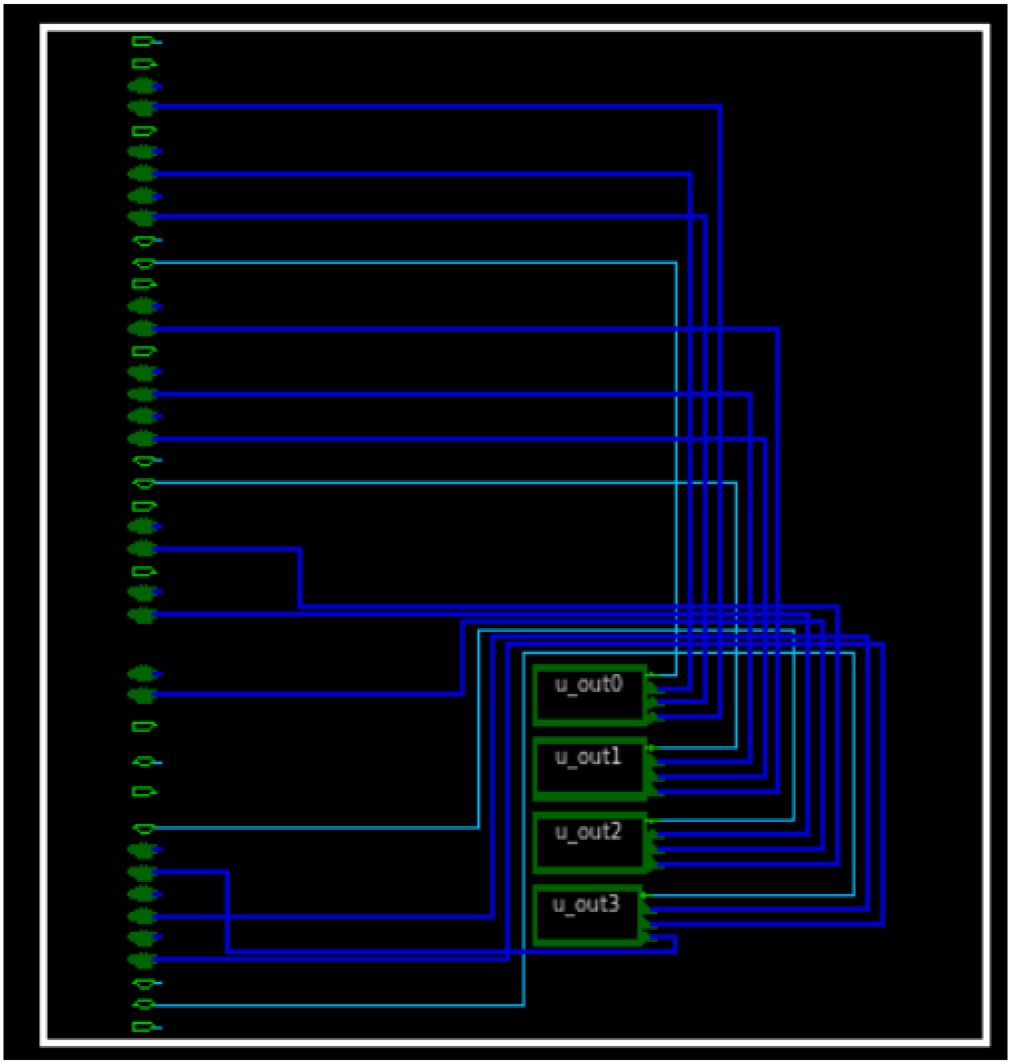
Simulation log in euclide is attached as simulation_log1.txt

Without clock gating

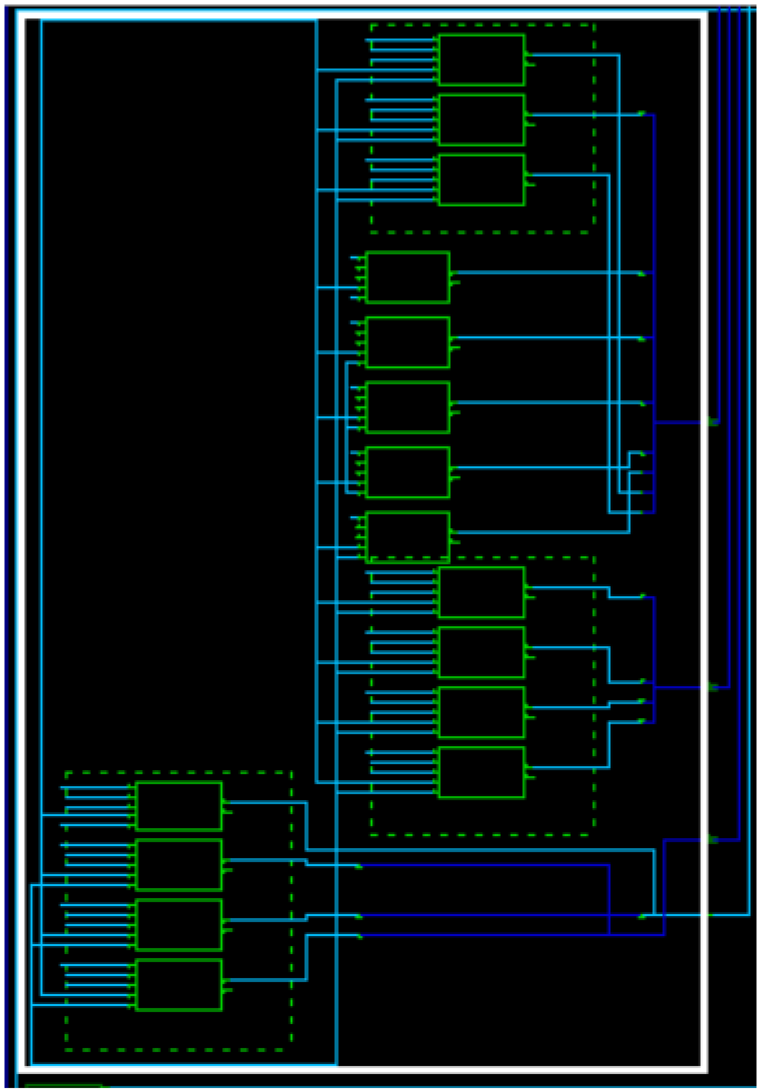
Layout

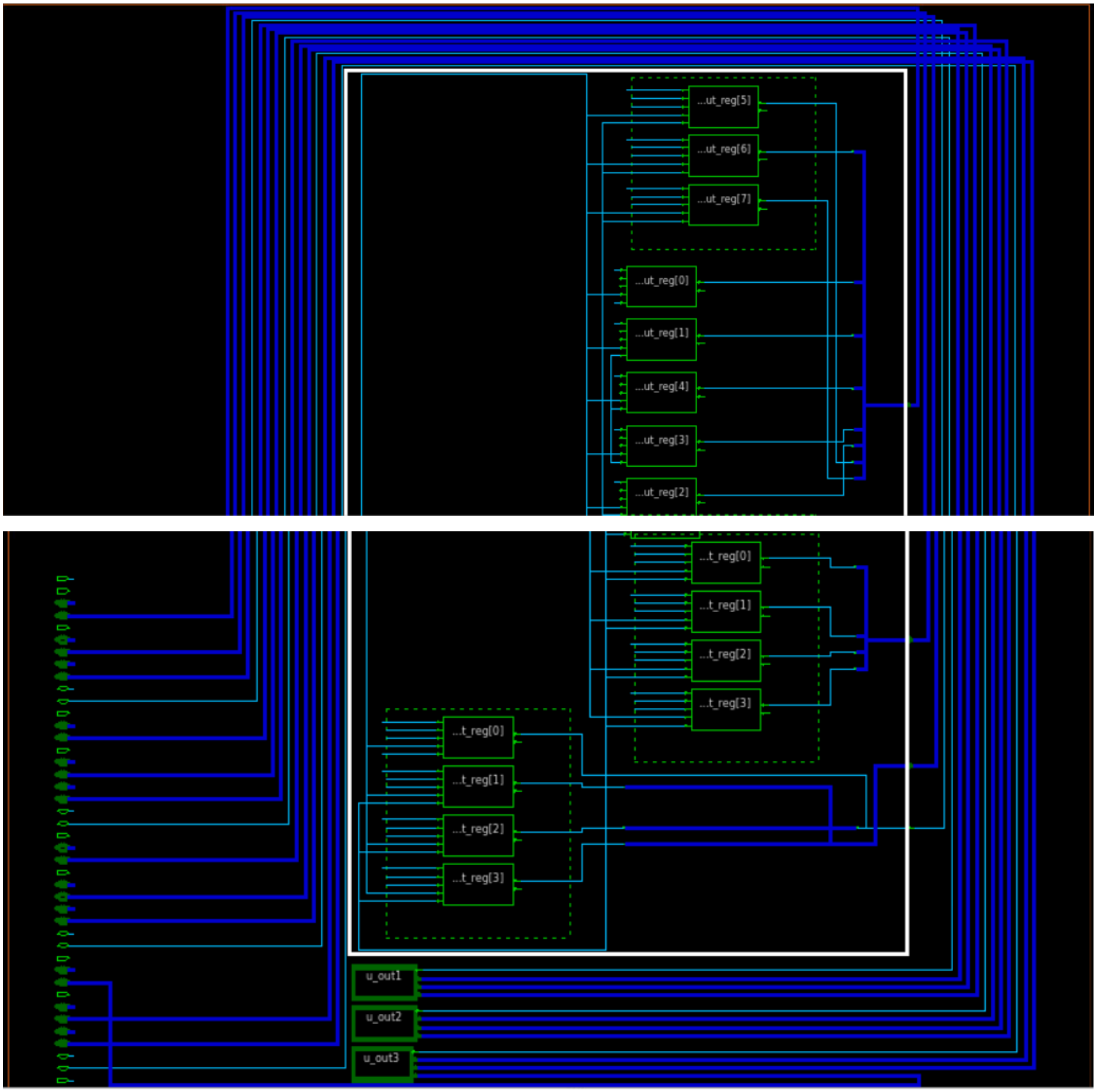


Schematic



Inside port





Gate level netlist is attached as switch_4port2.v

Timing report

fc_shell> report_timing
Information: Timer using 1 threads
Warning: Corner Slow: 0 process number, 3 process label, 0 voltage, and 0 temperature mismatches. (PVT-030)
Warning: 27199 cells affected for early, 27199 for late. (PVT-031)
Warning: 0 port driving_cells affected for early, 0 for late. (PVT-034)
Warning: Corner Fast: 0 process number, 3 process label, 0 voltage, and 0 temperature mismatches. (PVT-030)
Warning: 27199 cells affected for early, 27199 for late. (PVT-031)
Warning: 0 port driving_cells affected for early, 0 for late. (PVT-034)
Information: The stitching and editing of coupling caps is turned OFF for design 'switch2.dlib:switch2/final_opto.design'. (TIM-125)

```
Information: Design Average RC for design switch2  (NEX-011)
Information: r = 1.051250 ohm/um, via_r = 0.460670 ohm/cut, c = 0.078248 ff/um, cc = 0.000000 ff/um (X dir) (NEX-017)
Information: r = 1.384202 ohm/um, via_r = 0.578659 ohm/cut, c = 0.090794 ff/um, cc = 0.000000 ff/um (Y dir) (NEX-017)
Information: The RC mode used is VR for design 'switch_4port'. (NEX-022)
Information: Update timing completed net estimation for all the timing graph nets (TIM-111)
Information: Net estimation statistics: timing graph nets = 27317, routed nets = 0, across physical hierarchy nets = 0, parasitics cached nets = 27316, delay annotated nets = 0, parasitics annotated nets = 0, multi-voltage nets = 0. (TIM-112)
****************************************************************
Timer Settings:
Delay Calculation Style:                   auto
Signal Integrity Analysis:                 disabled
Timing Window Analysis:                    disabled
Advanced Waveform Propagation:             disabled
Variation Type:                            fixed_derate
Clock Reconvergence Pessimism Removal:     disabled
Advanced Receiver Model:                   disabled
LLE:                                       disabled
ML Acceleration:                           off
****************************************************************
****************************************
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -report_by design
Design : switch_4port
Version: V-2023.12-SP3
Date   : Sun Jan 18 16:11:36 2026
****************************************

  Startpoint: count_reg[2][1] (rising edge-triggered flip-flop clocked by clk)
  Endpoint: fifo_reg[2][63][target][1] (rising edge-triggered flip-flop clocked by clk)
  Mode: FUNC
  Corner: Slow
  Scenario: FUNC_Slow
  Path Group: clk
  Path Type: max

  Point                                            Incr      Path
  ------------------------------------------------------------------------
  clock clk (rise edge)                            0.00      0.00
  clock network delay (ideal)                      0.00      0.00

  count_reg[2][1]/CLK (SDFFARX1_HVT)               0.00      0.00 r
  count_reg[2][1]/Q (SDFFARX1_HVT)                 0.59      0.59 f
  ctmi_104957/Y (OR3X1_RVT)                        0.24      0.83 f
  phfnr_buf_8829/Y (INVX1_RVT)                     0.08      0.91 r
  ctmi_4053/Y (AND2X1_RVT)                         0.12      1.03 r
  phfnr_buf_8859/Y (INVX1_HVT)                     0.20      1.23 f
  ctmi_104955/Y (NOR2X1_RVT)                       0.31      1.54 r
  ctmi_104954/Y (NAND2X0_RVT)                      0.14      1.68 f
  ctmi_105163/Y (NAND2X0_RVT)                      0.16      1.84 r
  ctmi_105162/Y (AND3X4_RVT)                       0.28      2.12 r
  ctmi_105161/Y (OAI21X1_RVT)                      0.29      2.42 f
  ctmi_105160/Y (AND2X2_RVT)                       0.24      2.65 f
  ctmi_105159/Y (NAND2X1_RVT)                      0.27      2.92 r
  phfnr_buf_8927/Y (INVX2_RVT)                     0.12      3.04 f
  ctmi_105646/Y (NAND2X0_RVT)                      0.26      3.30 r
  ctmi_4060/Y (NOR2X0_RVT)                         0.23      3.53 f
  ctmi_105639/Y (NAND2X0_RVT)                      0.11      3.64 r
  phfnr_buf_9097/Y (INVX0_RVT)                     0.08      3.72 f
```

```
ctmi_105637/Y (NAND2X0_RVT)          0.12    3.84 r
phfnr_buf_9242/Y (INVX0_HVT)         0.25    4.10 f
ctmi_105636/Y (NAND2X0_RVT)          0.31    4.41 r
ctmi_4059/Y (NOR2X1_RVT)             0.22    4.62 f
phfnr_buf_9283/Y (INVX0_RVT)         0.08    4.71 r
ctmi_4058/Y (NAND3X4_RVT)            0.29    4.99 f
HFSINV_2144_12254/Y (INVX1_HVT)      0.19    5.18 r
HFSBUF_1958_12253/Y (NBUFFX8_HVT)    0.30    5.48 r
ctmi_108610/Y (AND2X4_RVT)           0.26    5.75 r
ctmi_108609/Y (NAND2X0_RVT)          0.21    5.95 f
phfnr_buf_9501/Y (INVX1_HVT)         0.25    6.20 r
ctmi_7584/Y (AO22X1_RVT)             0.28    6.48 r
phfnr_buf_9546/Y (INVX0_HVT)         0.26    6.74 f
ctmi_4066/Y (OR2X2_RVT)              0.35    7.10 f
HFSINV_427_11167/Y (INVX0_HVT)       0.16    7.25 r
HFSBUF_320_11166/Y (NBUFFX4_HVT)     0.27    7.52 r
ctmi_122659/Y (NAND2X2_RVT)          0.24    7.76 f
ctmi_122664/Y (NAND2X0_RVT)          0.20    7.97 r
phfnr_buf_9709/Y (INVX0_HVT)         0.32    8.29 f
ctmi_122663/Y (NAND4X0_RVT)          0.24    8.53 r
ctmi_122662/Y (NAND2X0_RVT)          0.16    8.70 f
ctmi_122661/Y (OAI222X1_RVT)         0.39    9.09 r
HFSINV_255_11145/Y (INVX2_RVT)       0.13    9.22 f
ctmi_122690/Y (AO22X1_RVT)           0.21    9.43 f
ctmi_122688/Y (AO221X1_RVT)          0.14    9.57 f
fifo_reg[2][63][target][1]/D (SDFFX1_RVT)    0.00    9.57 f
data arrival time                    9.57

clock clk (rise edge)                10.00   10.00
clock network delay (ideal)          0.02    10.02
fifo_reg[2][63][target][1]/CLK (SDFFX1_RVT)    0.00    10.02 r
clock uncertainty                    -0.15   9.87
library setup time                   -0.30   9.57
data required time                   9.57
---------------------------------------------------------------------------
data required time                   9.57
data arrival time                    -9.57
---------------------------------------------------------------------------
slack (MET)                          0.01


1
```

Power report

```
fc_shell> report_power
****************************************
Report : power
        -significant_digits 2
Design : switch_4port
Version: V-2023.12-SP3
Date   : Sun Jan 18 16:12:30 2026
****************************************
Information: Activity propagation will be performed for scenario FUNC_Slow.
Information: Doing activity propagation for mode 'FUNC' and corner 'Slow' with effort level 'medium'. (POW-024)
Information: Timer-derived activity data is cached on scenario FUNC_Slow (POW-052)
Infomation: Fast mode activity propagation power.rtl_activity_annotation setup is ignored. Always use accurate mode.
Information: Running switching activity propagation in scalar mode!

 **** Information : No. of simulation cycles = 6 ****
Mode: FUNC
```

Corner: Slow
Scenario: FUNC_Slow
Voltage: 0.75
Temperature: 125.00

Voltage Unit : 1V
Capacitance Unit : 1fF
Time Unit : 1ns
Temperature Unit : 1C
Dynamic Power Unit : 1pW
Leakage Power Unit : 1pW

Switched supply net power scaling:
scaling for leakage power

Supply nets:
VDD (power) probability 1.00 (default)
VSS (ground) probability 1.00 (default)
Warning: Power table extrapolation (extrapolation mode) for port D on cell count_reg[0][4] for parameter Tinp. Lowest table value = inf, highest table value = inf, value = 0.064602 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port CLK on cell count_reg[0][4] for parameter Tinp. Lowest table value = 0.016000, highest table value = 1.024000, value = 0.000000 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port Q on cell count_reg[0][4] for parameter Tinp. Lowest table value = 0.016000, highest table value = 1.024000, value = 0.000000 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port Q on cell count_reg[0][4] for parameter Cout. Lowest table value = 0.000100, highest table value = 0.008000, value = 0.000000 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port QN on cell count_reg[0][4] for parameter Tinp. Lowest table value = 0.016000, highest table value = 1.024000, value = 0.000000 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port D on cell count_reg[0][2] for parameter Tinp. Lowest table value = inf, highest table value = inf, value = 0.066032 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port CLK on cell count_reg[0][2] for parameter Tinp. Lowest table value = 0.016000, highest table value = 1.024000, value = 0.000000 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port Q on cell count_reg[0][2] for parameter Tinp. Lowest table value = 0.016000, highest table value = 1.024000, value = 0.000000 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port QN on cell count_reg[0][2] for parameter Tinp. Lowest table value = 0.016000, highest table value = 1.024000, value = 0.000000 (POW-046)
Warning: Power table extrapolation (extrapolation mode) for port D on cell u_out0/data_out_reg[6] for parameter Tinp. Lowest table value = inf, highest table value = inf, value = 0.148087 (POW-046)
Note - message 'POW-046' limit (10) exceeded. Remainder will be suppressed.
Warning: Fall toggles on pin count_reg[0][6]/QN are impossible given input states; converted to rise toggles. (POW-069)
Warning: Fall toggles on pin count_reg[1][6]/QN are impossible given input states; converted to rise toggles. (POW-069)
Warning: Fall toggles on pin count_reg[3][6]/QN are impossible given input states; converted to rise toggles. (POW-069)

Cell Internal Power = 1.66e+09 pW (94.3%)
Net Switching Power = 1.00e+08 pW (5.7%)
Total Dynamic Power = 1.76e+09 pW (100.0%)

Cell Leakage Power = 1.68e+09 pW

Attributes
----
u - User defined power group
i - Includes clock pin internal power

Power Group	Internal Power	Switching Power	Leakage Power	Total Power	(%)	Attrs
io_pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	(0.0%)	
memory	0.00e+00	0.00e+00	0.00e+00	0.00e+00	(0.0%)	
black_box	0.00e+00	0.00e+00	0.00e+00	0.00e+00	(0.0%)	
clock_network	1.42e+09	0.00e+00	0.00e+00	1.42e+09	(41.3%)	i

```
register              1.30e+08          2.71e+06          7.63e+08          8.95e+08    ( 26.1%)
sequential            0.00e+00          0.00e+00          0.00e+00          0.00e+00    (  0.0%)
combinational         1.09e+08          9.74e+07          9.14e+08          1.12e+09    ( 32.6%)
--------------------------------------------------------------------------------------------------------------------------------
Total              1.66e+09 pW       1.00e+08 pW       1.68e+09 pW       3.43e+09 pW
1
```

Area report

```
fc_shell> report_area
****************************************
Report : area
Design : switch_4port
Version: V-2023.12-SP3
Date   : Sun Jan 18 16:13:18 2026
****************************************

Number of ports:                  305
Number of nets:                 27472
Number of cells:                27199
Number of combinational cells:  23219
Number of sequential cells:      3980
Number of macros/black boxes:       0
Number of buf/inv:               1895
Number of references:             118

Combinational area:          61740.47
Buf/Inv area:                 3712.28
Noncombinational area:       34474.13
Macro/Black Box area:            0.00

Total cell area:             96214.60
1
```

Utilization report

```
fc_shell> report_utilization
****************************************
Report : report_utilization
Design : switch_4port
Version: V-2023.12-SP3
Date   : Sun Jan 18 16:13:42 2026
****************************************
Utilization Ratio:                  0.7218
Utilization options:
 - Area calculation based on:           site_row of block switch2/final_opto
 - Categories of objects excluded:      hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                         133300.5612
Total Capacity Area:                133300.5612
Total Area of cells:                96214.5980
Area of excluded objects:
 - hard_macros         :            0.0000
 - macro_keepouts      :            0.0000
 - soft_macros         :            0.0000
 - io_cells            :            0.0000
 - hard_blockages      :            0.0000
Total Area of excluded objects:     0.0000
Ratio of excluded objects:          0.0000
```

Utilization of site-rows with:
 - Site 'unit':   0.7218

0.7218

Clock gating report

```
fc_shell> report_clock_gating
****************************************
Report : clock_gating
Design : switch2
Version: V-2023.12-SP3
Date   : Sun Jan 18 16:14:11 2026
****************************************


                   Clock Gating Summary
    ------------------------------------------------------------------
    |  Number of Clock gating elements         |            0       |
    |                                          |                    |
    |  Number of Tool-Inserted Clock gates     |         0   (0.00%) |
    |                                          |                    |
    |  Number of Pre-Existing Clock gates      |         0   (0.00%) |
    |                                          |                    |
    |  Number of Gated registers               |         0   (0.00%) |
    |                                          |                    |
    |  Number of Tool-Inserted Gated registers |         0   (0.00%) |
    |                                          |                    |
    |  Number of Pre-Existing Gated registers  |         0   (0.00%) |
    |                                          |                    |
    |  Number of Ungated registers             |      3980 (100.00%) |
    |                                          |                    |
    |  Total number of registers               |          3980      |
    |                                          |                    |
    |  Max number of Clock Gate Levels         |            0       |
    ------------------------------------------------------------------


1
```

Synthesis script

Switch.sdc

Same file as previously

Run.tcl

```
#lappend search_path scripts design_data
set_host_options -max_cores 4
set TECH_FILE     "/data/synopsys/lib/saed32nm/ref/tech/saed32nm_1p9m.tf"
##########################################################################
### Physical Library Settings
##########################################################################
create_lib  -technology $TECH_FILE  -ref_libs { /data/synopsys/lib/saed32nm/ref/CLIBs/saed32_hvt.ndm
/data/synopsys/lib/saed32nm/ref/CLIBs/saed32_lvt.ndm /data/synopsys/lib/saed32nm/ref/CLIBs/saed32_rvt.ndm }  switch2.dlib
open_lib switch2.dlib
report_ref_libs
read_parasitic_tech -tlu /data/synopsys/lib/saed32nm/ref/tech/saed32nm_1p9m_Cmax.lv.tluplus -name Cmax
read_parasitic_tech -tlu /data/synopsys/lib/saed32nm/ref/tech/saed32nm_1p9m_Cmin.lv.tluplus -name Cmin
```

```
#
save_lib
analyze -format sverilog {port_if.sv switch_port.sv switch_4port.sv}
elaborate switch_4port
set_top_module switch_4port
# save hir
set_app_options -name  compile.flow.autoungroup  -value false
#start_gui
save_block -as switch2/elaborate

# mcmm_setup:
# Remove all MCMM related info
remove_corners   -all
remove_modes     -all
remove_scenarios -all
# Create Corners
create_corner Fast
create_corner Slow
#
## Set parasitics parameters
set_parasitics_parameters -early_spec Cmin -late_spec  Cmin -corners {Fast}
set_parasitics_parameters -early_spec Cmax -late_spec  Cmax -corners {Slow}
#
## Create Mode
create_mode  FUNC
current_mode FUNC
#
## Create Scenarios
create_scenario -mode FUNC -corner Fast    -name FUNC_Fast
create_scenario -mode FUNC -corner Slow    -name FUNC_Slow
#

#sourse ConFiles/con431.con
current_scenario FUNC_Fast
source  /project/verif/users/mariandawud/ws/portsC/switch.sdc
current_scenario FUNC_Slow
source  /project/verif/users/mariandawud/ws/portsC/switch.sdc

set_auto_floorplan_constraints -core_utilization 0.7 -side_ratio {1 1} -core_offset 2

set_clock_gating_enable -exclude [all_registers]
set_clock_gating_options -minimum_bitwidth 1000000 -max_fanout 1000000

compile_fusion -to logic_opto
#create_placement
#legalize_placement
##Power
compile_fusion -to final_opto

save_block -as switch2/final_opto
report_timing

#set_attribute [get_layers {M1 M3 M5 M7 M9}]   routing_direction vertical
#set_attribute [get_layers {M2 M4 M6 M8 MRDL}] routing_direction horizontal

#source create_pg_network.tcl
#clock_opt
#route_opt
#save_block -as ALU/route_opt
write_sdf switch2.sdf
Write_verilog switch_4port2.v
```

Simulation log in euclide is attached as simulation_log2.txt